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Subject code & Name :	ITUE301: Advanced Web Development Frameworks	Practical :	11	Academic Year :	2026-27

Task - 11

Aim

To containerize the full-stack web application using Docker and Docker Compose, designing a multi-stage Dockerfile for the frontend SPA served via Nginx, a lightweight Node.js backend container, an isolated bridge network, and persistent named volumes for MongoDB data retention.

Code

GitHub Link / Repository Link: https://github.com/Dharmikvaghani/AWDF-PRACTICAL.git

Key Implementation Highlights:

1. Multi-stage frontend Dockerfile: Node.js 20 build stage -> Nginx Alpine production runtime.
2. Backend Dockerfile: Node.js 20-alpine with production dependency caching and 0.0.0.0 binding.
3. docker-compose.yml: Coordinates frontend (8080), backend (5000), and mongodb (27017) over app-network.

```
// docker-compose.yml
services:
  mongodb:
    image: mongo:7.0
    volumes: [ awdf-mongodb-data-practical11:/data/db ]
  backend:
    build: ./backend
    environment:
      MONGO_URI: mongodb://mongodb:27017/taskdb_docker
    depends_on: [ mongodb ]
  frontend:
    build: ./frontend
    ports: [ "8080:80" ]
    depends_on: [ backend ]
Output
```

Output

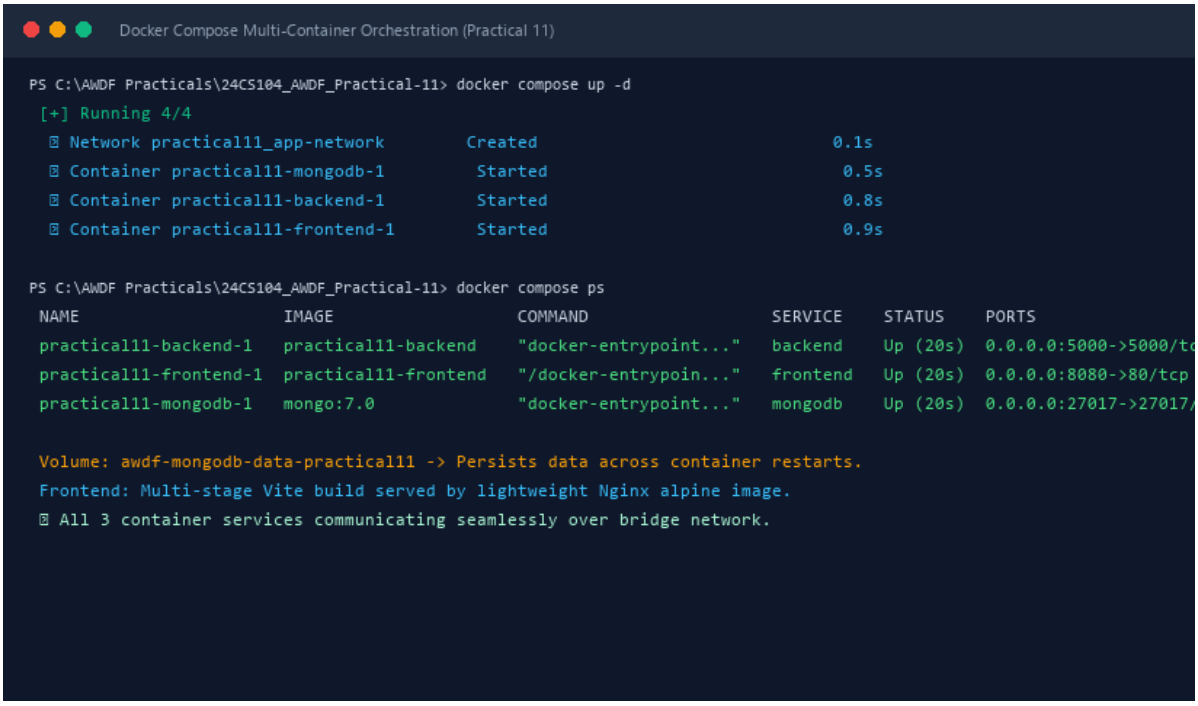


Figure 11: Execution Output & Functional Verification for Practical 11